Acadian (ACD) Variant Overview of the Forest Vegetation Simulator

DRAFT variant overview / General Technical Report scaffold. 29 June 2026, updated 7 July 2026.

Structured after the USDA Forest Service Northeast (NE) Variant Overview (FVS Staff 2008, rev. August 2023) so it can become the official FVS-ACD variant documentation. Equation content is drawn from the Acadian GY Model Documentation (Weiskittel and team, 2025-05-20) and the AcadianGY source, built on the 12.7.1 development line. AcadianGY 12.10.1 is the current operational and production version, documented here. Sections still marked [POPULATE] need team-written text before publication (the geographic range map figure, the keyword and database usage reference, the full Literature Cited list, and the Appendix C/D validation and keyword material); the species and component parameter tables (Appendices A and B) are populated directly from the AcadianGY source and are unchanged from v12.9.0 through v12.10.1, since the later refinements touch mortality, ingrowth, self-thinning, and the management modifiers, not the species-level growth, crown, taper, or volume parameters. Plaintext math, no em dashes; this draft will be rendered to a formatted Word/PDF GTR. Updated 7 July 2026 to document v12.10.1 as the operational version, with the version lineage through 12.5.0, 12.7.0/12.7.1, 12.9.x, and 12.10.1, the five-way independent scorecard and honest scorecard position, the Bakuzis biological-realism pass, the disclosed ingrowth limit, and the planned v13 program.

Authors and Contributors

Aaron Weiskittel and the University of Maine Cooperative Forestry Research Unit (CFRU) developed the Acadian growth-and-yield model. Ben Rice (Midgard Natural Resources) maintains the FVS-ACD implementation and benchmarking. Component equations are credited to their authors throughout (Kuehne, Russell, Rijal, Li, Chen, Castle, Cen, and others; see Literature Cited). This overview documents the FVS-ACD variant as implemented in AcadianGY 12.10.1, the current operational version.

Suggested citation: Weiskittel, A., Rice, B., and contributors. 2026. Acadian (ACD) Variant Overview - Forest Vegetation Simulator. CFRU / University of Maine, Orono, ME. [pages].

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Parameter	Default
Projection cycle length	1 year (FVS reporting frequency user-selectable)
Region code	ME (northern New England); also NB, NS, QC
Site productivity	Climate Site Index (CSI, m); modern ESI accepted in the same input (12.6.x)
Maximum stand density index	Composition-based, Weiskittel and Kuehne (2019); see 3.5
Species	37 Acadian species codes plus OH (other hardwood) and OS (other softwood)

Minimum DBH

2.5 cm for height prediction; ingrowth threshold DBH user-set

Units

Metric internally (DBH cm, height m, BA m2/ha, TPH per ha)

Volume

Kozak variable-exponent taper; Westfall and Scott

1. Introduction

The Acadian forest region spans northern New England and Atlantic Canada. It is a transitional mixed conifer and hardwood type, sitting between the northern hardwood forests to the south and the boreal forest to the north (Loo and Ives 2003). The Acadian growth-and-yield model, developed by the CFRU team led by Aaron Weiskittel, is the regionally specific model for this setting, and FVS-ACD is its implementation within the Forest Vegetation Simulator framework. It departs from the FVS Northeast (NE) variant in two respects. Its core growth, mortality, height, crown, ingrowth, and taper relationships were fit on Acadian data, with the Kuehne et al. (2022) increment equations alone drawing on more than 2.5 million diameter and a million height observations, and it is driven by a climate-based site index in place of a traditional site index. FVS-ACD is a one-year-cycle, individual-tree, distance-independent model and, like the other variants, supports the FVS extensions.

2. Geographic Range

The Acadian variant applies to northern New England (Maine, New Hampshire, Vermont) and Atlantic Canada (New Brunswick, Nova Scotia) plus southern Quebec, the region for which the component equations were fit. Region codes select the regional mortality and modifier parameters (ME, NB, NS, QC). [POPULATE: range map figure, analogous to FVSne figure 2.0.1.]

3. Control Variables

3.1 Region codes

ME (default, northern New England), NB (New Brunswick), NS (Nova Scotia), QC (Quebec). Region selects the regional parameter sets for stand and tree mortality and the spruce budworm modifier.

3.2 Species codes

37 rows are parameterized in sp.attr (35 distinct named species plus the OH other-hardwood and OS other-softwood composite groups, and a 99 fallback row). Species not in the parameter set map to OH or OS by softwood/hardwood type. Full species code table with scientific names, FIA and PLANTS codes, specific gravity, and shade/drought/waterlog tolerance in Appendix A; component parameter tables (height, HCB, crown width, bark ratio, mortality, modifiers, ingrowth, taper, volume) by sp.grp.ingrowth and by individual species in Appendix B.

3.3 Habitat / ecological unit

FVS-ACD does not use habitat type codes; site productivity enters through the climate site index (3.4). Level 3 ecoregion is available as an optional stratifier for maximum density (3.5).

3.4 Site index

The Acadian model uses Climate Site Index (CSI, m), a climate-and-soil-driven productivity surface (Weiskittel et al. 2011), fit by random forest on seven climatic variables (most important: mean temperature of the warmest month, the warmest-to-coldest month temperature difference, and the growing-season-to-annual precipitation ratio). CSI is a user input and is available as raster surfaces for current and future (2030, 2060, 2090) climate normals. CSI enters total height, diameter increment, stand basal-area increment, height increment, mortality, ingrowth, and maximum SDI.

Operational note (12.7.0). The production model uses the validated modern Ecological Site Index (ESI) as the site driver, supplied through the same input slot. ESI tracks measured site index well (R^2 0.91). A controlled full-set test showed that ESI and the legacy climate site index give the same regional volume and value, with an all-region basal-area bias of +1.4 percent under both. The ESI gain therefore does not appear in volume or value bias. It appears in dominant-height behavior and biological realism, since ESI anchors the stand dominant height onto a fitted Chapman-Richards curve (k = 0.0039, p = 0.726, base age 50; section 4.1 note), which removes the Bakuzis height site-ordering inversions. The ESI surface is supplied as a raster (Acadian ESI v8). The earlier AcadianSIm climate surfaces are a separate future-climate product and are not the operational site driver.

3.5 Maximum density (maximum SDI)

Stand density index is additive at the plot level, SDI = sum over trees of (DBH/25.4)^1.6 * TPH, with DBH in cm and TPH per ha. Maximum SDI follows Weiskittel and Kuehne (2019), a composition- and structure-based function:

SDImax = 475.2079 - 1.5908*HW.ba.percent - 236.9051*log(meanSG) + 50.3299*sqrt(DBHrange)
+ 13.5202*spp.div + 0.0685*Elev - 2.8537*sqrt(Elev) + 222.7836*(1/CSI)

where HW.ba.percent is percent hardwood basal area, meanSG is basal-area-weighted mean specific gravity (capped at 0.80), DBHrange is the plot DBH range, spp.div is species count, and Elev is elevation (m). A fallback SDImax2 = 1347.445 - 1003.870*meanSG is used where the primary function is undefined. Relative density is SDI/SDImax.

Version 12.6.x note: a brms re-estimation of plot-level SDImax across northern New England found SDImax is essentially independent of site index (correlation with CSI and ESI near 0) and is primarily a function of forest type (softwood ceilings near 1050 to 1100, northern hardwood near 900, wet hardwood near 750), which the composition terms above already approximate through mean specific gravity. A site-independent SDImax (dropping the 222.78/CSI term, recentred) is available behind a flag.

Version 12.9.0 note: a CONUS-scale brms re-estimation (independent of the northern New England one above) confirmed forest type dominates the variance in plot-level maximum SDI (R-squared 0.21) far more than site index alone (R-squared 0.02), and that a localized, forest-type-and-ecoregion-aware maximum predicts observed self-thinning roughly twice as well as a generic species-table constant. Applied to AcadianGY, the diagnostic finding was that the Weiskittel-Kuehne SDImax equation above already handles composition well through the percent-hardwood and specific-gravity terms, but under-corrects specifically for hardwood-dominated stands in Quebec outside the Acadian-core ecoregion (CEC North American Level III: Acadian Plains and Hills, Maritime Lowlands). v12.9.0 adds a gated multiplier to SDImax (ops$sdimax.fortype.tight.mult, default 0.65 to 0.70), active only when a stand is

simultaneously Quebec, outside the Acadian core, and hardwood-dominated by BA-weighted dominant species group. See section 10 and 20260701_v129_full_validation.md for the full derivation and validation.

4. Growth Relationships

4.1 Height-diameter (total tree height)

HTPred, after Rijal et al. (2012a) modified for additional species:

ht = 1.37 + ((c0 + c0.spp) + CSI^c1) * (1 - exp(-c2*DBH))^(c3 + c3.spp + c4*log(CCF+1) + c5*BAL)

CSI enters the asymptote (the CSI^c1 term). c0 to c5 are fixed; c0.spp and c3.spp are species random effects. Used to fill missing or invalid heights. Species parameter table in Appendix B.1.

4.2 Bark ratio

DOBtoDIB (Weiskittel and Li 2012) converts diameter outside to inside bark. Form and species parameters in Appendix B.4.

4.3 Crown ratio and crown recession

Height to crown base, HCBPred (Rijal et al. 2012b, refit with species random effect, Weiskittel et al. 2013):

hcb = HT / (1 + exp((a0 + a0.spp) + a1*DBH + a2*HT + a3*DHR + a4*log(CCF+1) + a5*(BAL+1)))

Crown recession, dHCB (Russell et al. 2014):

dHCB = (CL + dHT) / (1 + exp(b1 + b2*log(CR+0.01) + b3*log(CCF+1) + b4*log(1.01-CR)
+ b5*shade^2 + b6*log(shade*CR)))

HCB species random-effect table (a0.spp) in Appendix B.2. The crown-recession equation uses fixed parameters for all species (no separate species table in the source); shade tolerance is read from Appendix A.

4.4 Crown width

Maximum crown width mcw = a1*DBH^a2 (open-grown, Russell and Weiskittel 2011, 99th-quantile); largest crown width lcw = mcw / (b1*DBH^b2) for forested conditions. Species parameters in Appendix B.3.

4.5 Crown competition factor

CCF is the sum of tree maximum crown areas per plot, used in height, height-to-crown-base, crown recession, increment, and ingrowth.

4.6 Diameter increment

dDBH, Kuehne et al. (2022), species as a random effect, fit on > 2.5 million observations:

dDBH = exp((b0+b0.SPP) + b1*log(DBH) + (b2+b2.SPP)*DBH + (b3+b3.SPP)*log(CR)
+ (b4+b4.SPP)*log(BAL.SW+0.1) + b5*BAL.HW + b8*log(CSI))

Fixed-effect coefficients (Kuehne et al. 2022, Table 6): b0 = -1.64234, b1 (ln DBH) = 0.376978, b2 (DBH) = -0.025684, b3 (ln CR) = 0.713456, b4 (ln BAL.SW+0.1) = -0.065750, b5 (BAL.HW) = -0.017740, b8 (ln CSI) = 0.135377. Species random effects apply to the intercept, DBH, ln(CR), and ln(BAL.SW+0.1). Competition is one-sided and asymmetric (basal area in larger softwoods and hardwoods); two-sided metrics (SDI, relative density) were no more effective. On independent FIA data the equation is nearly unbiased (mean bias -1.2 percent) and improves on the prior FVS-ACD diameter and basal-area increment submodels by 11 to 13 percent in mean absolute bias. Tree diameter increment is constrained by the stand basal-area increment (4.8). Full species random-effect tables in Kuehne et al. (2022) Table S3.

4.7 Height increment

dHT, Kuehne et al. (2022), more conservative than the earlier Russell et al. (2014) form (addresses the FVS-NE over-prediction expectation):

dHT = exp((b0+b0.SPP) + b1*log(HT) + (b2+b2.SPP)*HT + b3*CR + b4*(CCFL/100) + b7*CSI^2)

Fixed-effect coefficients (Kuehne et al. 2022, Table 8): b0 = -2.19445, b1 (ln HT) = 0.426404, b2 (HT) = -0.064710, b3 (CR) = 0.394837, b4 (CCFL/100) = -0.011430, b7 (CSI^2) = 0.000294. Species random effects apply to the intercept and HT. The new dHT improves on the prior Russell et al. (2014) submodel substantially (mean absolute bias down 41 to 74 percent), the largest accuracy gain of the two increment refits. Full species random-effect tables in Kuehne et al. (2022) Table S5.

4.8 Stand basal-area increment

Chen et al. (in prep.), added at v12.3.0, applied at the stand level as a cap on summed tree diameter increment:

dBA = rt*Ba*(1 - Ba/k); rt = q0 + q1*log(RD+1e-6) + q2*log(CSI) + q3*log(pHW.ba*100+1e-6)
+ q4*log(CSI*RD+1e-6)

Applied only when summed predicted tree BA increment exceeds the stand prediction, and capped at 0.227 percent of plot BA (5th percentile, Maine FIA). RD is relative density SDI/SDImax. This is the density-dependent brake on growth.

5. Mortality Model

Stand- and tree-level mortality, Weiskittel et al. (2012). The multi-stage approach of Chen et al. (2023) was evaluated in beta and was not superior, so the model uses the 2012 formulation (see the v12 history memo).

Stand mortality probability: prob = exp(k)/(1+exp(k)), k = b0 + b1*BA* + *b2*BA^2 + b3*BAG* + *b4*QMD + b5*BA.BF* + *b6*BA.IH. Stand mortality basal area: BA.mort = (b0 + b1*pBA.bf* + *b2*pBA.ih)*BA^(b3* + *b4*BAG); BF.mort = b5*BA.BF^b6* + *b7*(QMD.BF/QMD). Tree survival: logistic in DBH and DBH^2. Stand mortality is distributed across trees in proportion to tree mortality probability. Region-specific parameters (ME, NB, NS, QC) in Appendix B.6; tree-level background mortality species parameters (the DBH/DBH^2 survival logistic and the Weibull-form diameter-jump term) in Appendix B.5.

Operational note: the production model uses the 2012 stand and tree mortality with SDImax-ceiling self-thinning (ops

$use.sdimort)active,adensity-frontiermortalitythatholdspost-stepSDIatorbelowSI$
use.mortcal) is available but trades stem-count accuracy and ships off. The principled next-version fix is a regional refit of the 2012 stand-mortality coefficients.

Coupled self-thinning and density-released ingrowth (12.7.1). The 12.7.0 self-thinning used a single slope above a relative-density onset, which on long-horizon projection never plateaued (basal area climbed past 36 m2/ha and was still rising at age 120). Version 12.7.1 replaces it with a coupled two-segment relative-density mortality. Let RD = SDI/SDImax. Mortality is zero below RD 0.30, rises on a gentle slope from 0.30 to a break at 0.62 (ops$sdimort.onset,opss$dimort.break, ops $sdimort.beta.gentle),andrisesonasteepslopeabovethebreak(opss$dimort.beta.steep), so a fully stocked stand self-thins along the SDImax line. It is coupled with a density-released ingrowth (section 6). Through a logistic in RD, recruitment is suppressed at high RD and released below about RD 0.58. The biological rationale is the build, self-thin, and rebound dynamic of a stand that grows to its maximum, self-thins along the density frontier, and rebounds as ingrowth releases when density falls, the process underlying a shifting-mosaic steady state. On a 160-year projection of a young dense spruce-fir stand across the site gradient, basal area builds and plateaus at about 33 to 34 m2/ha by mid-rotation and then levels, inside the realistic fully-stocked Acadian spruce-fir range (about 28 to 34), replacing the 12.7.0 runaway climb. The decline-and-rebound is critically damped at the single-stand deterministic scale. Because mature Quebec-edge stands and moderately dense New Brunswick stands overlap in relative density, this relative-density lever reduces the mature-stand over-prediction while modestly deepening the growing-stand under-prediction; separating the two is the next-version regional and composition-aware mortality target.

Regional and composition-aware SDImax gate (12.9.0). Closer diagnosis showed that the 12.7.1 mature-stand residual is not a Quebec-wide phenomenon. It concentrates in hardwood-dominated Quebec stands outside the Acadian-core ecoregion, whereas softwood-dominated Quebec stands and all of New Brunswick were already close to unbiased. v12.9.0 adds a gated multiplier to the SDImax term used by this self-thinning mechanism (section 3.5), active only when a stand is simultaneously province Quebec, ecoregion outside the Acadian core, and hardwood-dominated. This raises the effective self-thinning intensity for that specific population and closes the target cell's total-volume bias from +56.7 to +10.4 percent, with no effect on New Brunswick or on the independent recent Maine FIA growing-stand check (see section 10). The mechanism adds to the coupled self-thinning above rather than replacing it, scaling SDImax before the RD = SDI/SDImax ratio that drives the two-segment mortality curve.

6. Regeneration and Ingrowth

Full ingrowth model, Li et al. (2011), for ten species groups (BCH birch, BF, RM, SPR spruce, WP, OH, AB, QA, SM, WC). A hurdle model: occurrence probability P(occur) and count E[count] = exp(eta), both functions of BA, percent hardwood, TPH, CSI, minimum DBH, and QMD. The occurrence is gated by a cut-point (CutPoint, default 0.95). FVS-ACD also retains FVS resprouting after simulated harvest. Composition and occurrence-probability parameters by species group in Appendix B.8.

Version 12.6.1 note: the 0.95 occurrence gate suppresses recruitment on most plot-years and is the source of the under-recruitment behind the stem-count deficit. v12.6.1 makes the statistically correct expected-value recruitment the default, E[ingrowth] = P(occur) * E[count] (set CutPoint to 0), with a calibration multiplier. This is the dominant lever on stem count and QMD but, by global sensitivity analysis, does not materially change stand value (the recruits are small).

Version 12.7.1 note (density-released ingrowth): observed recruitment, tree-ID matched on the raw MagPlot records, is about 40 stems per ha per year in New Brunswick and about 20 in Quebec. v12.7.1 couples the expected-value recruitment to relative density through a logistic in RD (ops $use.ingrowth.release, ops$ingrowth.rd.center 0.58, ops$ingrowth.rd.k): recruitment is suppressed at high relative density and released below about RD 0.58. Coupled with the two-segment self-thinning mortality (section 5), this produces the build, self-thin, and rebound stand dynamic.

7. Volume and Taper

Stem taper: Kozak variable-exponent taper and Westfall and Scott (Li et al. 2012, Weiskittel and Li 2012). Bark thickness via DOBtoDIB. Merchantable volume and product specifications follow the FVS volume framework; for value work, sawlog (International 1/4-inch board feet) and pulp (cords) are split by species and size. Kozak and Westfall-Scott taper species parameters in Appendix B.9, legacy Honer volume species parameters in Appendix B.10, bark ratio in Appendix B.4.

Operational note (12.7.x): all volume and biomass in the benchmarking and validation are reported on the FIA National Scale Volume and Biomass (NSVB) system, applied through an emulator that predicts the NSVB responses (gross cubic volume, net merchantable volume, net sawlog volume, and aboveground dry biomass) from log(DIA squared times HT) by species group with Duan smearing, with R-squared 0.96 to 0.99 against the FIA NSVB outputs. This puts observed and predicted tree lists on a single, current volume-and-biomass currency and replaces the earlier single form-factor merchantable volume used for increment scoring.

8. Modifiers

Thinning (Kuehne et al. 2016; note the source code header instead credits Weiskittel, Hennigar, and Kershaw 2015 for this modifier, a citation discrepancy to reconcile before publication): diameter growth, mortality, height growth, and crown recession modifiers keyed on removal intensity and timing. Spruce budworm (Cen et al. 2018): balsam fir and spruce mortality, diameter and height growth, keyed on cumulative defoliation, region ME and NB. Hardwood form and risk (Castle et al. 2018): mortality and diameter growth modifiers for selected hardwoods. All parameter tables in Appendix B.7.

Direct-effect management modifiers (12.10.1). The current operational version adds the Chapter 11 direct-effect management modifiers as gated additive releases on planted and precommercial-thinning (PCT) stands. The direct effect is expressed as a robust additive term on the class of per-stand observed minus predicted-untreated increment, in place of a diameter-growth multiplier pinned to a guard clamp (planted +3.83 $m^2\ ha^{-1}$ basal area, PCT +2.41 $m^2\ ha^{-1}$, with proportional releases in the other metrics). It is gated by ops$use.mgmt.mod and applies only to planted and PCT stands, so the gate-off run is byte identical to v12.9.x on unmanaged stands. The release is a diameter-growth term and not a mortality term, is physically bounded within the unmanaged density envelope, and does not break the self-thinning plateau. See section 10.3 for the fit and validation.

9. Model Application

CSI is supplied by the user or sampled from the raster surfaces (current and 2030/2060/2090 climate normals); recent testing finds the 2030 surface reasonable. Cycle length is one year. The model is run from a tree list through the FVS interface or the AcadianGY R driver (make_stand, make_acd_tree,

make_ops, AcadianGYOneStand). [POPULATE: keyword and database usage, analogous to Essential FVS.]

10. Version lineage, current operational version (12.10.1), and validation

AcadianGY 12.10.1 is the current operational and production version, frozen as the operational release for the 18 August 2026 Acadian Growth and Yield meeting. Version 13 is in progress, but no v13 version is gated, so nothing in the v13 program below is a claimed result. v12.10.1 is the endpoint of a single development line that began with 12.5.0 and added one gated refinement at a time. Each refinement is individually isolated, so that with its flag in the legacy position the build reproduces the prior version bit for bit. The operational configuration is the Kuehne et al. (2022) diameter and height increment equations (4.6, 4.7) as the growth base; the validated ESI v8 as the site driver with the fitted ESI-consistent dominant-height anchor recalibration active (3.4, 4.1), which delivers the clean Bakuzis site ordering (10.2); a further top-height anchor-level parity correction available behind a gate but shipping off in the current config, deferred to v13.0 (see 10.3); FIA NSVB volume and biomass (7); expected-value, density-released ingrowth (6); the coupled two-segment relative-density self-thinning (5); the regional and composition-aware SDImax gate for the Quebec hardwood edge (5, 3.5); and the Chapter 11 direct-effect management modifiers as gated additive releases on planted and precommercial-thinning stands (8, and section 10.3 below).

The version lineage, in order of development:

12.5.0, the prior operational and production base. Kuehne et al. (2022) increment equations, climate site index, and the 2012 stand and tree mortality. On long-horizon projection it does not self-thin, with basal area climbing past 62 $m^2\ ha^{-1}$ at age 120, and is not recommended for mature-stand or long-horizon use.

12.7.0 and 12.7.1. The 12.7.x line added the validated ESI v8 site driver and the ESI-anchored dominant height, FIA NSVB volume and biomass through the emulator (R-squared 0.96 to 0.99 against FIA NSVB), the coupled two-segment relative-density self-thinning, and the density-released ingrowth. Together these make the long-horizon basal-area trajectory plateau at the biologically correct Acadian maximum (about 33 to 34 $m^2\ ha^{-1}$) rather than running away, couple recruitment to relative density so that a stand builds, self-thins along the density frontier, and rebounds as ingrowth releases, and put observed and predicted tree lists on a single current volume and biomass currency.

12.9.x. A CONUS-scale brms re-estimation of plot-level maximum SDI (independent of the earlier northern New England one) confirmed that composition and ecoregion, not site index, dominate the variance in the self-thinning ceiling. v12.9.x adds a gated multiplier to SDImax active only when a stand is simultaneously Quebec, outside the Acadian-core ecoregion, and hardwood-dominated. This closes the remaining mature-stand over-prediction in exactly that population, with an independent recent-FIA cross-check confirming zero effect on New Brunswick or Maine (the province term in the gate excludes Maine by construction).

12.10.1. Chapter 11 direct-effect management modifiers, expressed as gated additive releases on planted and precommercial-thinning (PCT) stands (section 10.3). The modifier is gated by ops$use.mgmt.mod and touches only planted and PCT stands; the gate-off run is byte identical to v12.9.x on the 266 unmanaged stands. This is the current operational version.

Every refinement is gated, and with the flags in their legacy position the build reproduces the base bit for bit, so each is individually isolated and validated. v12.10.1 with its management gate off reproduces v12.9.x exactly, and v12.9.x with its own gate unset reproduces 12.7.1 exactly.

Provenance note on the operational build. The operational scorecard baseline documented in this section was produced from the AcadianGY_12.9.2.r source plus the recorded operational ops recipe, the gated management release applied through make_ops on the v12.9.x engine. A standalone AcadianGY_12.10.1.r script that folds that recipe into the source now exists in the repository and carries the AcadianV12.10.1 version tag; it is being reconciled against the recipe-driven baseline to confirm it reproduces the scored results byte for byte before it supersedes the recipe as the canonical operational file.

10.1 Four-model independent scorecard (1 July 2026)

A multi-dimensional scorecard (quantitative bias, RMSE, modeling efficiency, and periodic annual increment; qualitative Bakuzis biological realism, long-horizon trajectory shape, transparency, data requirements, usability, and uncertainty characterization) was run on AcadianGY 12.5.0, 12.7.0, 12.7.1, and the current OSM-ACD release (v2.26.1, confirmed current against the OSM-Contributors GitHub organization and the Forus Research download page), on identical independent in-domain data. The full scorecard is 20260701_four_model_scorecard.md; figures five_model_scorecard_heatmap.png and five_model_scorecard_radar.png (v12.9.0 added as a fifth model, section 10.2).

The headline finding is that OSM-ACD carries the lowest pooled level bias of the four models on every metric and, once it is scored on all nine NSVB metrics through the same emulator harness as FVS-ACD, also carries the highest pooled modeling efficiency. An earlier read that showed a negative Nash-Sutcliffe efficiency for OSM on every metric was an index-join artifact. OSM output had been matched to the observed tree lists on a mismatched key rather than reflecting a real lack of per-stand skill. Rescoring OSM through the common NSVB harness resolves it, and OSM's pooled modeling efficiency is positive and leads the set. FVS-ACD leads OSM on biological realism, the climate-ready site driver, and quadratic mean diameter, not on pooled level bias or modeling efficiency. All three AcadianGY versions explain 55 to 78 percent of out-of-sample variance on every metric. AcadianGY 12.5.0 fails the Bakuzis self-thinning check outright, with long-horizon basal area still climbing past 62 m2/ha at age 120 and several thousand stems per hectare still alive, and is not recommended for mature-stand or long-horizon use. Of the FVS-ACD versions, AcadianGY 12.7.1 has the best combined quantitative and qualitative profile prior to the 12.9.0 refinement below.

10.2 v12.9.0: a targeted SDImax correction for the Quebec hardwood edge

The scorecard in 10.1 localized 12.7.1's remaining structural gap to a specific population: stands that are simultaneously Quebec (province), outside the Acadian-core ecoregion (CEC North American Level III: Acadian Plains and Hills, Maritime Lowlands are the core; Northern Appalachian and Atlantic Maritime Highlands, Algonquin and Southern Laurentians, Central Laurentians and Mecatina Plateau, and Eastern Great Lakes Lowlands are outside it), and hardwood-dominated (BA-weighted dominant species group sugar maple, birch, aspen, other hardwood, or American beech). Softwood-dominated Quebec stands, and every New Brunswick stand regardless of composition, were already close to unbiased and are untouched by the v12.9.0 gate.

Development process, briefly (full derivation 20260701_v129_full_validation.md, 20260701_v129_ooda_stress_test.md): a region-only multiplier was tried first and found to be a pure

trade-off (fixing basal area worsened volume and vice versa, with no setting fixing both). An ESI-based spatial cluster was tried and found weak, because ESI v8 coverage of Quebec is patchy. Forest type (BA-weighted dominant species group) located the real signal. True ecoregion sharpened it further. An ecoregion-only gate (no province term) closed the Quebec target as well as the final version, but an independent cross-check against the recent Maine FIA growing-stand stratum caught a real regression: the correction leaked onto Maine's own hardwood stands sharing the same ecoregion as the Quebec problem stands, worsening the FIA annual-increment check from -52 to -83 to -93 percent. Restoring province as a third gate dimension eliminated this by construction, since no Maine plot can ever satisfy a Quebec-only gate.

Validation, in-domain MagPlot (n = 733) and independent recent Maine FIA (n = 788), predicted versus observed t2 level bias:

Target cell (Quebec, outside Acadian core, hardwood-dominated, n = 163): total volume +56.7 to +10.4 percent, stand value to +4.4 percent. Basal area, already under-predicted in 12.7.1, deepens further (-9.3 to -18.1 percent), the SDImax-driven trade-off confined now to exactly the population needing it.

New Brunswick: unchanged from 12.7.1 in every metric.

Recent Maine FIA: unchanged from 12.7.1 (basal-area level -2.2 percent, annual increment -0.085 m2/ha/yr), confirming the gate cannot affect Maine plots.

A stand-level split-half robustness check (163 target-cell stands, two random halves) confirmed the correction is not driven by a handful of outlier stands.

Known residual: a seeded 160-year trajectory of the corrected hardwood population plateaus at about 21 m2/ha, below the realistic 28 to 32 m2/ha target for that composition (the same composition, uncorrected, plateaus at 31 to 32). A correction calibrated on a 5 to 15 year remeasurement window, carried through a full rotation, under-shoots the long-horizon asymptote. v12.9.0 is recommended for short-to-mid rotation inventory projection; v12.7.1 remains the recommended version for carbon and old-growth applications on this specific population until a tapered version of the correction, relaxing toward the true asymptote rather than applying a flat multiplier throughout the rotation, is fit.

Independent validation, and what carries to the frozen harness. This subsection is the development-era 12.7.0 to 12.7.1 self-thinning validation. It was run on a non-frozen dominant-height calibration and on populations, recent Maine FIA (n=788) and the ESI-covered Quebec edge (n=378), that are not in the frozen per-stand MagPlot files behind the section 10.3 five-way, so it has not been, and cannot cheaply be, re-benched on the frozen operational harness. Its results split by whether they depend on the dominant-height anchor factor (afac). A code-level check of the frozen harness confirms the afac enters at one point only, the dominant-height increment multiplier (dht.mult, operationally NB 0.579 and QC 0.111), and never touches the diameter-increment path or the density-dependent mortality that sets trees per hectare and quadratic mean diameter. The self-thinning results reported here are therefore density-frontier quantities that do not depend on the afac and carry to the frozen harness unchanged: the Reineke slopes, the plateau-versus-runaway trajectory finding, and the 12.7.0 to 12.7.1 self-thinning improvement. The level-bias figures that follow, and any dominant-height or NSVB-volume number in particular, were computed at the development afac and are not frozen-consistent; they are the original developmental record, not frozen results. The authoritative frozen scorecard is the five-way in section 10.3.

On genuinely independent, in-domain data, the full FIA NSVB metric set, predicted versus observed t2 level bias, New Brunswick (n=355) and the ESI-covered Quebec edge (n=378), 12.7.0 to 12.7.1:

New Brunswick total volume -3.9 to -9.8 percent, biomass -1.4 to -7.3, value -4.5 to -10.2; basal area -3.5 to -8.5.

Quebec edge total volume +23.3 to +14.4 percent, biomass +19.1 to +10.2, value +27.3 to +18.3; basal area -2.1 to -9.3.

Top-height (dominant-height) bias is deliberately not tabulated here. It is the single metric the afac scales directly, so the development-era figures ran warm and are not frozen-consistent. On the frozen operational harness the dominant-height afac (NB 0.579, QC 0.111) brings every build's dominant height into the tight +3.7 to +4.6 percent band reported for the five-way in section 10.3; that band, not the development-era top-height numbers, is the operational result.

The coupled self-thinning reduces the mature-stand Quebec-edge volume and biomass over-prediction by roughly a third to a half. New Brunswick and Quebec stands overlap in relative density, so the same relative-density mortality that self-thins the dense Quebec stands also deepens the moderately dense New Brunswick under-prediction; the two are ends of one lever and are the next-version regional and composition-aware mortality target.

Recent Maine FIA (post-2010, independent in time, n=788): basal-area level bias +0.7 percent in 12.7.0 to -2.2 percent in 12.7.1 (still near-unbiased); annual basal-area increment +11 percent (over) in 12.7.0 to 52 percent under (still positive net growth) in 12.7.1. On independent New Brunswick MagPlot, the v12.7 generation is +0.05 m3/ha/yr on a shared form-factor volume harness, near zero and at least as good as the 2018 OSM-ACD (-0.23) and FVS-ACD v10.6 (-0.18); the 2018 conclusion that OSM clearly beat FVS-ACD does not hold for v12.7 on independent New Brunswick data.

Biological realism (Bakuzis matrix): site ordering clean (0 of 125 inversions, the dominant-height anchor), Reineke self-thinning slopes -1.28 to -1.35, one moderate Eichhorn flag. Observed ingrowth about 40 stems per ha per year in New Brunswick and 20 in Quebec, matched by the density-released form. The Reineke self-thinning slopes are a property of the density frontier, the slope of log(TPH) on log(QMD), and as noted above do not depend on the dominant-height afac, so they stand on the frozen harness unchanged.

10.3 v12.10.1: direct-effect management modifiers, the current operational version

v12.10.1 adds the Chapter 11 direct-effect management modifiers as gated additive releases on planted and precommercial-thinning (PCT) stands, the current operational version. Relative to the v12.10.0 prototype, the planted and PCT diameter-growth multipliers no longer pin to the old 0.7 to 1.6 guard clamp. The direct effect is re-expressed as a robust additive release, planted +3.83 $m^2\ ha^{-1}$ basal area and PCT +2.41 $m^2\ ha^{-1}$ (ten percent trimmed means over the class of per-stand observed minus predicted-untreated increment, with proportional releases in the other metrics), which is equivalent to a net multiplier of about 3.0 planted and 2.3 PCT, above the old 1.6 ceiling, which is why the clamped version still under-predicted planted. The modifier is gated by ops$use.mgmt.mod and applies only to planted and PCT stands; the gate-off run is byte identical to v12.9.x on the 266 unmanaged stands. The additive fold coefficients move with the data (planted basal-area release 3.85 to 4.40, PCT 2.45 to 2.76 across cross-validation folds), so it is a fitted release, not a clamped constant. It is physically bounded: managed

predicted SDI stays inside the unmanaged envelope, the release is a diameter-growth term rather than a mortality term, and the self-thinning plateau holds at every amplitude tested.

The honest scorecard position. A five-way independent benchmark (v12.5.0, v12.7.0, v12.9.x, v12.10.1, and OSM v2.26.1) on the identical held-out in-domain data (MagPlot New Brunswick n=355 and ESI-covered Quebec n=350, plus recent Maine FIA for the growing-stand cross-check) places v12.10.1 level with v12.9.x on the pooled operational domain, both behind OSM there, and ahead of both on the managed cell. Because the frozen operational harness fully specifies the operational ops, v12.7.1, v12.9.x, and v12.10.1 are byte identical on the pooled base scorecard and differ only in the managed cells; the five-way therefore resolves to three distinct pooled engines, v12.5.0, v12.7.0, and the shared v12.7.1, v12.9.x, and v12.10.1 operational line.

Table 10.1. Pooled operational scorecard, v12.10.1 versus the comparison models (nine NSVB metrics).

model

mean absolute level bias (%)

mean modeling efficiency

TOST level equivalences (of 9)

v12.5.0

7.1

0.71

4

v12.7.0

8.0

0.73

2

v12.9.x

4.9

0.71

7

v12.10.1

4.9

0.71

7

OSM v2.26.1

3.2

0.75

8

OSM keeps the best pooled level bias (3.2 versus 4.9), the best mean modeling efficiency (0.75 versus 0.71), and the most TOST level equivalences (8 versus 7 of 9, against four of nine for v12.5.0 and two of nine for v12.7.0). On the frozen harness every model's top height lands in a tight +3.7 to +4.6 percent band, so the apparent top-height over-prediction once read into the older 12.5.0 and 12.7.0 lines was an artifact of the pre-freeze basis and does not survive re-benching; the two older engines separate instead on volume, v12.5.0 a strong total-volume over-predictor (+9.3 percent, mean absolute level bias 7.1) and v12.7.0 a volume under-predictor (-8.3 percent, mean absolute level bias 8.0). v12.10.1 does not claim pooled parity. The point of v12.10.1 is the managed cell. The Chapter 11 gate closes the managed gap sharply in absolute basal-area bias, planted from -24.9 to -4.4 percent, PCT from -15.6 to -1.3 percent, and the managed class as a whole from -19.0 to -2.5 percent, and v12.10.1 beats OSM on every managed class in absolute basal-area bias (managed 2.5 versus 11.2, planted 4.4 versus 20.9, PCT 1.3 versus 5.4). One frozen-basis caveat to flag: gate-on planted total volume now sits at -10.8 percent, because the frozen anchor lowers untreated planted-stand volume so the fixed additive release closes less of the volume gap; the basal-area releases themselves are unchanged. On the pooled domain v12.10.1 and v12.9.x are identical to one decimal, because the modifier touches only managed stands and those are a small share of the in-domain pool, so pooled mean absolute bias (4.9), RMSE, and modeling efficiency all carry over from v12.9.x unchanged and the modifier's effect appears only in the managed cell. On the pooled domain the FVS-ACD advantage is quadratic mean diameter (about +1 percent versus OSM +11 percent), biological realism, transparency, and a climate-ready site driver, not pooled precision.

Biological realism (Bakuzis pass). v12.10.1 inherits the v12.9.x self-thinning machinery and, on seeded 160-year projections across the ESI gradient, passes all five relations on the modified Bakuzis matrix (Leary 1997): Eichhorn (volume a function of height), Reineke self-thinning slope (fitted -1.59 against a -1.6 target), site ordering with no overtaking, the yield-density plateau (about 36 $m^2\ ha^{-1}$), and Sukachev (size versus density). With the management modifier on, a seeded planted-softwood projection still

plateaus at every amplitude tested (standard deviation of basal area over the last 20 years about 0.21, flat), and the ceiling rises smoothly with amplitude because basal area at the Reineke limit rises with tree size. By contrast v12.5.0 fails self-thinning outright in the same seeded battery. OSM is not scored in this seeded-gradient battery and enters the comparison only through the matched-stand trajectory, so the Bakuzis pass is a statement about FVS-ACD's seeded-gradient behavior rather than a five-way ranking.

Disclosed ingrowth limit. Observed MagPlot recruitment, tree-id matched across remeasurements, is 39.8 stems ha^{-1} yr^{-1} in New Brunswick and 19.7 in Quebec. The shipped expected-value ingrowth hurdle predicts 17.9 and 11.7 stems ha^{-1} yr^{-1}, about 45 percent of observed in New Brunswick and 59 percent in Quebec. Recruitment responds in the correct direction to density and management (planted and PCT stands, which carry lower residual density, recruit more than unmanaged), and the expected-value hurdle lifted recruitment well off the near-zero floor of the classic 2018 too-infrequent FVS-ACD problem, but a residual under-prediction of the absolute level remains. This is neither the 2018 near-zero failure nor an inflated recruitment; it is a real, moderate under-prediction, disclosed as a limit and a target for the v13 ingrowth refit. It does not change the shipped version and does not touch the managed-cell scorecard, which is a volume and basal-area result on the tag threshold.

Pooled parity was tested and deferred. A top-height anchor-level correction was tested as a reach for pooled parity with OSM and deferred to v13.0. In the shipped config a top-height over-prediction of about +4 percent only partly offsets a basal-area under-prediction of about -10 percent, so FVS-ACD total volume sits modestly under observed on the unmanaged core, about -4 percent, rather than near zero. Correcting top height alone removes the offsetting positive term and drives total volume further below observed rather than toward zero, widening rather than closing the pooled mean absolute bias gap to OSM. True parity therefore needs a joint re-anchor of dominant height and basal area together, recorded as the v13.0 domain-plus-increment refit. The gated lever exists in the source (neutral factor reproduces v12.10.1) but is not adopted for the operational version.

10.4 Planned v13

The v13 program addresses the disclosed limits above rather than the operational cell v12.10.1 already wins. It moves from gated levers on fixed equations to refitted, site-driven, region-and-composition-aware component equations with uncertainty, each step gated by the same multi-dimensional scorecard against the prior version and OSM. The program is planned in four releases and none is gated as of this writing, so nothing below is a result. It is the roadmap, not a claim.

The first release, v13.0, is a joint dominant-height and basal-area re-anchor. The model is masked to its calibrated Acadian domain and dominant height and the increment are re-anchored together on the expanded remeasurement data, so the top-height over-prediction and the basal-area under-prediction are corrected in one step rather than left to partly offset each other. This addresses the partial error cancellation that currently leaves total volume modestly under observed on the unmanaged core (section 10.3), which the one-sided top-height lever tested this cycle showed cannot be corrected alone without driving volume further negative.

The second release, v13.1, is a Bayesian hierarchical refit of mortality and ingrowth with region, ecoregion, and forest-type random effects and the brms maximum-SDI posterior as the self-thinning asymptote. Fitting the two components as a system with shared random effects estimates directly the regional and composition structure now carried by the gated SDImax multiplier and the two-segment relative-density knob, so the growing and mature tails become separable by construction, and it targets the

disclosed ingrowth limit (section 10.3) by matching observed recruitment (about 40 stems ha^{-1} yr^{-1} in New Brunswick, 20 in Quebec) rather than under-predicting it. The hierarchical fits carry parameter and residual uncertainty into prediction intervals, a scorecard dimension v12 does not satisfy.

The third release, v13.2, carries management as a growth-scale release estimated per tree, so the additive stand-level release of v12.10.1 (section 10.3) is reproduced through the full engine by a fitted per-tree amplitude rather than a post-hoc net release applied at the stand level.

The fourth release, v13.3, adds a climate-scenario mode driven by the current and future site-productivity surfaces, kept separate from the validated base so that projections can be run under alternative climate normals without ever benchmarking on a projected surface.

Until v13 is gated, v12.10.1 is the recommended operational version for inventory projection, with v12.7.1 retained for long-horizon carbon and old-growth work on the specific hardwood population where the v12.9.x flat multiplier under-shoots the realistic plateau.

11. Literature Cited

[POPULATE with full references: Loo and Ives 2003; Weiskittel et al. 2011, 2012, 2013; Weiskittel and Li 2012; Weiskittel and Kuehne 2019; Rijal et al. 2012a,b; Russell and Weiskittel 2011; Russell et al. 2014; Kuehne et al. 2016, 2022; Li et al. 2011, 2012; Chen et al. (in prep., 2023); Cen et al. 2018; Castle et al. 2018; Jiang et al. 2014; Leary 1997 (Bakuzis); Dixon 2002 (Essential FVS).]

12. Appendices

Appendices A and B below are populated directly from the AcadianGY 12.9.0 source (scripts/AcadianGY_12.9.0.r), extracted programmatically from the working parameter tables so the values match the running model exactly. Appendix C (keyword reference) and Appendix D (validation tables and figures) are not yet populated: C needs team-written usage documentation analogous to the Essential FVS keyword reference (Dixon 2002), and D should pull the final tables and figures directly from the companion benchmarking report (FVS-ACD_12.9_benchmark_report.md) once that report is finalized, rather than duplicating numbers here. Formatting note: several Appendix B tables have 9 to 12 columns and render narrow in a portrait Word layout (see rendered docx); the final typeset GTR should set these in landscape orientation or a smaller table font rather than leaving them word-wrapped as in this draft.

Appendix A. Species Code and Attribute Table

Alpha species codes as parameterized in AcadianGY 12.9.0 (sp.attr), with common and scientific names, FIA species code, PLANTS symbol, type (HW = hardwood, SW = softwood), specific gravity (sg), green wood density (wd, g per cm3), and shade, drought, and waterlog tolerance scores (0 to 5 scale, internal Acadian model compilation). sp.grp.ingrowth is the crown, ingrowth, and background-mortality

parameter group the species maps to in Appendix B. 37 rows total: 35 distinct named species plus the OH (other hardwood) and OS (other softwood) composite groups, and a final 99 fallback row used when an input species code matches none of the above. RN and PC are retained aliases, not distinct species: RN is the FVS-NE alpha code for red pine (identical parameters to RP), and PC and PR are both carried as pin cherry with identical parameters, an inherited duplicate kept in the source table for input compatibility.

Code

Common name

Scientific name

FIA code

PLANTS

Type

sg

wd

Shade

Drought

Waterlog

sp.grp.ingrowth

AB

American beech

Fagus grandifolia

531

FAGR

HW

0.64

0.56

4.75

1.5

1.5

AB

AS

Ash (unspecified)

Fraxinus spp.

540

FRAXI

HW

0.57

0.51

2.84

2.74

3.02

OH

BA

Black ash

Fraxinus nigra

543

FRNI

HW

0.5

0.45

2.96

2

3.5

OH

BC

Black cherry

Prunus serotina

762

PRSE2

HW

0.5

0.47

2.46

3.02

1.06

OH

BF

Balsam fir

Abies balsamea

12

ABBA

SW

0.35

0.33

5.01

1

2

BF

BP

Balsam poplar

Populus balsamifera

741

POBA2

HW

0.34

0.31

1.27

1.77

2.63

OH

BS

Black spruce

Picea mariana

95

PIMA

SW

0.46

0.38

4.08

2

2

SPR

BT

Bigtooth aspen

Populus grandidentata

743

POGR4

HW

0.39

0.36

1.21

2.5

2

OH

EC

Eastern cottonwood

Populus deltoides

742

PODE3

HW

0.4

0.37

1.76

1.57

3.03

OH

EH

Eastern hemlock

Tsuga canadensis

261

TSCA

SW

0.4

0.38

4.83

1

1.25

OS

GA

Green ash

Fraxinus pennsylvanica

544

FRPE

HW

0.56

0.53

3.11

3.85

2.98

OH

GB

Gray birch

Betula populifolia

379

BEPO

HW

0.48

0.45

1.5

2.34

1

BCH

HH

Eastern hophornbeam

Ostrya virginiana

701

OSVI

HW

0.78

0.63

4.58

3.25

1.07

OH

JP

Jack pine

Pinus banksiana

105

PIBA2

SW

0.43

0.4

1.36

4

1

OS

NS

Norway spruce

Picea abies

91

PIAB

SW

0.43

0.3702

4.45

1.75

1.22

OS

OH

Other hardwood (group)

n/a (composite group)

998

n/a

HW

0.5121

0

2.29

0

0

OH

OS

Other softwood (group)

n/a (composite group)

298

n/a

SW

0.445

0

2.27

0

0

OS

PB

Paper birch

Betula papyrifera

375

BEPA

HW

0.55

0.48

1.54

2.02

1.25

BCH

PC

Pin cherry [alias of PR]

Prunus pensylvanica

761

PRPE2

HW

0.38

0.36

2.26

0

0

OH

PR

Pin cherry

Prunus pensylvanica

761

PRPE2

HW

0.38

0.36

2.26

0

0

OH

QA

Quaking aspen

Populus tremuloides

746

POTR5

HW

0.38

0.35

1.21

1.77

1.77

QA

RB

River birch

Betula nigra

373

BENI

HW

0.62

0.49

1.45

1.53

2.85

BCH

RM

Red maple

Acer rubrum

316

ACRU

HW

0.54

0.49

3.44

1.84

3.08

RM

RP

Red pine

Pinus resinosa

125

PIRE

SW

0.46

0.41

1.89

3

1

OS

RN

Red pine [FVS-NE alias of RP]

Pinus resinosa

125

PIRE

SW

0.46

0.41

1.89

3

1

OS

RO

Northern red oak

Quercus rubra

833

QURU

HW

0.63

0.56

2.75

2.88

1.12

OH

RS

Red spruce

Picea rubens

97

PIRU

SW

0.4

0.37

4.39

2.5

2

SPR

SB

Sweet birch

Betula lenta

372

BELE

HW

0.65

0.6

2.58

3

1

BCH

SM

Sugar maple

Acer saccharum

318

ACSA3

HW

0.63

0.56

4.76

2.25

1.09

SM

ST

Striped maple

Acer pensylvanicum

315

ACPE

HW

0.46

0.44

3.56

2

1

OH

TA

Tamarack (eastern larch)

Larix laricina

71

LALA

SW

0.53

0.49

0.98

2

3

OS

WA

White ash

Fraxinus americana

541

FRAM2

HW

0.6

0.55

2.46

2.38

2.59

OH

WC

Northern white cedar

Thuja occidentalis

241

THOC2

SW

0.31

0.29

3.45

2.71

1.46

WC

WP

Eastern white pine

Pinus strobus

129

PIST

SW

0.35

0.34

3.21

2.29

1.03

WP

WS

White spruce

Picea glauca

94

PIGL

SW

0.4

0.33

4.15

2.88

1.02

SPR

YB

Yellow birch

Betula alleghaniensis

371

BEAL2

HW

0.62

0.55

3.17

3

2

BCH

99

Other (unmatched-species fallback)

n/a

n/a

n/a

HW

0.3

0.3

3

0

0

OH

Specific gravity and green wood density source Miles and Smith (2009), USDA FS-NRS-38, as compiled in the Acadian parameter set.

Group	Description
AB	American beech group
BF	Balsam fir
BCH	Birch group (paper, gray, river, sweet, yellow)
OH	Other hardwood (composite)
OS	Other softwood (composite)
QA	Quaking aspen

RM

Red maple

SM

Sugar maple

SPR

Spruce group (black, red, white)

WC

Northern white cedar

WP

Eastern white pine

Appendix B. Component Parameter Tables

All tables below are extracted directly from the AcadianGY 12.9.0 R source (scripts/AcadianGY_12.9.0.r) and correspond one-to-one with the working parameter objects the model uses at runtime (sp.attr, mcw.spp, ht.pred.spp, etc., named in each subsection). Species-level diameter increment and height increment random effects (section 4.6, 4.7) are not reproduced here; the fixed-effect coefficients are given in the main text and the full species random-effect tables are published in Kuehne et al. (2022) Tables S3 and S5.

B.1 Height-diameter species parameters (section 4.1, ht.pred.spp)

Species random-effect terms c0.spp and c3.spp added to the fixed c0 and c3 in the total height equation (Rijal et al. 2012a); nomatch defaults to OH. Fixed parameters: c0 = 12.44847305, c1 = 0.801705832, c2 = 0.043617034, c3 = 1.048674338, c4 = 0.011483716, c5 = -0.007550999.

Code

Common name

c0.spp

c3.spp

AB

American beech

-1.633

-0.1238

AE

American elm

-0.692

0.03461

AH

American hornbeam

-5.98

-0.03278

AI

Ailanthus (tree-of-heaven)

-6.45

-0.0984

AP

Apple spp.

-12.07

-0.476

AS

Ash (unspecified)

6.698

-0.1253

BA

Black ash

-1.617

-0.1416

BC

Black cherry

-4.527

-0.1726

BE

Boxelder

-1.606

-0.4246

BF

Balsam fir

1.775

0.1572

BL

Black willow

-9.828

-0.2926

BN

Butternut

0.8612

-0.1032

BO

Black oak

1.17

-0.04312

BP

Balsam poplar

2.522

-0.06336

BR

Bur oak

5.443

0.3544

BS

Black spruce

3.886

0.1887

BT

Bigtooth aspen

5.283

-0.06706

BW

American basswood

2.525

0.1539

EC

Eastern cottonwood

7.088

0.1821

EH

Eastern hemlock

0.06435

0.2607

EL

Elm spp. (other)

-0.4602

0.1942

GA

Green ash

-1.473

-0.07439

GB

Gray birch

-1.039

-0.2387

HH

Eastern hophornbeam

-2.454

-0.1764

HK

Hackberry

-0.1392

-0.1071

HT

Hackberry/hickory (unresolved short code)

-0.4984

0.0005085

JP

Jack pine

12.2

0.5071

NC

Non-commercial hardwood / northern white cedar fallback (context-dependent)

-0.1548

0.05067

NS

Norway spruce

10.76

0.7615

OH

Other hardwood (group)

-0.1523

-0.07739

OP

Other pine species

-1.252

0.197

OS

Other softwood (group)

6.644

0.155

PB

Paper birch

2.856

-0.05313

PI

Other spruce species

6.119

0.3962

PL

Pin oak (short-code variant, unresolved)

-12.58

-0.3544

PP

Pitch pine

-2.035

0.02845

PR

Pin cherry

-5.279

-0.2767

PY

Swamp cottonwood

2.4

-0.03028

QA

Quaking aspen

5.285

-0.01669

RC

Eastern redcedar

-13.36

-0.365

RL

Slippery elm

-13.46

-0.3733

RM

Red maple

1.134

-0.1249

RN

Red pine [FVS-NE alias of RP]

2.354

0.4332

RO

Northern red oak

0.5672

5.492e-05

RS

Red spruce

3.289

0.1978

SB

Sweet birch

5.059

0.2633

SC

Scotch pine (short-code variant of SP)

-1.777

0.273

SE

Sourwood (unresolved short code, context sweetgum/sourwood group)

-1.204

-0.2175

SH

Shagbark hickory

3.424

0.008524

SM

Sugar maple

1.831

-0.1509

SO

Scarlet oak

-0.3376

0.002666

ST

Striped maple

-4.215

-0.1585

SV

Silver maple

-1.668

-0.1804

SW

Swamp white oak (short-code variant)

1.308

0.03166

TA

Tamarack (eastern larch)

3.039

-0.07014

WA

White ash

1.586

-0.1526

WC

Northern white cedar

-2.638

0.1571

WO

White oak

-0.1796

0.05001

WP

Eastern white pine

1.892

0.2179

WS

White spruce

2.831

0.2849

YB

Yellow birch

-1.135

-0.1796

B.2 Height-to-crown-base species parameters (section 4.3, hcb.pred.spp)

Species random-effect term a0.spp added to the fixed intercept in the HCB equation (Rijal et al. 2012b, refit with species random effect per Weiskittel et al. 2013); nomatch defaults to OH. Fixed parameters: a0 = 0.29070, a1 = 0.00636, a2 = -0.02288, a3 = 0.08232, a4 = -0.03086, a5 = -0.01701. The crown-recession equation (dHCB, Russell et al. 2014) uses the same fixed parameters for all species (no species-specific table in the source); shade tolerance from Appendix A enters the shade term directly.

Code

Common name

a0.spp

AB

American beech

-0.2184

AI

Ailanthus (tree-of-heaven)

0.08171

AP

Apple spp.

0.1775

BA

Black ash

0.1127

BC

Black cherry

-0.1988

BF

Balsam fir

0.09359

BP

Balsam poplar

-0.1367

BR

Bur oak

0.0156

BS

Black spruce

-0.2278

BT

Bigtooth aspen

0.01004

EC

Eastern cottonwood

0.5073

EH

Eastern hemlock

0.4039

GB

Gray birch

-0.1816

HK

Hackberry

0.1146

JP

Jack pine

-0.2708

NC

Non-commercial hardwood / northern white cedar fallback (context-dependent)

-0.2051

NS

Norway spruce

0.9556

OH

Other hardwood (group)

-0.04289

OP

Other pine species

-0.718

PB

Paper birch

-0.1809

PI

Other spruce species

0.5486

PR

Pin cherry

-0.2709

QA

Quaking aspen

0.06083

RM

Red maple

-0.2025

RN

Red pine [FVS-NE alias of RP]

-0.1505

RO

Northern red oak

-0.3702

RS

Red spruce

-0.1213

SC

Scotch pine (short-code variant of SP)

0.2442

SE

Sourwood (unresolved short code, context sweetgum/sourwood group)

0.243

SM

Sugar maple

-0.1638

TA

Tamarack (eastern larch)

-0.304

WA

White ash

-0.1483

WC

Northern white cedar

0.4196

WO

White oak

0.1672

WP

Eastern white pine

-0.1467

WS

White spruce

0.1003

YB

Yellow birch

0.003235

B.3 Crown width species parameters (section 4.4, mcw.spp, lcw.spp)

Maximum crown width mcw = a1 * DBH^a2 (open-grown, 99th-quantile) and largest crown width lcw = mcw / (b1 * DBH^b2), Russell and Weiskittel (2011), 15 species plus NC, OH, OS, and a 99 fallback. RB shares BF, GB, and RO's grouped coefficients where noted; nomatch defaults to 99.

Code

Common name

a1 (mcw)

a2 (mcw)

b1 (lcw)

b2 (lcw)

BF

Balsam fir

1.37

0.572

1.49

0.105

BS

Black spruce

0.535

0.742

1

0.174

EH

Eastern hemlock

2.44

0.408

1.9

-0.057

WP

Eastern white pine

1.24

0.585

1

0.147

NC

Non-commercial hardwood / northern white cedar fallback (context-dependent)

1.63

0.436

2.19

-0.08

RS

Red spruce

1.8

0.461

4.33

-0.264

WS

White spruce

1.5

0.496

2.09

-0.069

AB

American beech

2.93

0.434

1

0.194

GB

Gray birch

2.24

0.382

3.1

-0.214

RB

River birch

2.24

0.382

3.1

-0.214

RO

Northern red oak

4.08

0.31

4.1

-0.272

PB

Paper birch

1.48

0.623

2.1

-0.035

QA

Quaking aspen

1.31

0.586

2.65

0.157

RM

Red maple

2.17

0.491

2.63

-0.132

SM

Sugar maple

3.31

0.356

1

0.161

YB

Yellow birch

4.04

0.308

4.23

-0.264

OH

Other hardwood (group)

4.04

0.308

2.65

0.157

OS

Other softwood (group)

1.597

0.514

2.328

0.02784

99

Other (unmatched-species fallback)

2.243

0.4627

2.793

-0.09011

B.4 Bark ratio / DOB-to-DIB species parameters (section 4.2, bark.ratio.spp)

Weiskittel and Li (2012). Where pcnt.bark > 0, DIB = DOB * (1 - pcnt.bark/100) (fixed percent-bark species); otherwise DIB = b0 * DOB^b1.

Code

Common name

pcnt.bark

b0

b1

AB

American beech

7

1

1

BC

Black cherry

10

1

1

BF

Balsam fir

0

0.878

1.025

BP

Balsam poplar

0

0.8737

1.012

BA

Black ash

0

0.8499

1.041

BS

Black spruce

0

0.871

1.026

BT

Bigtooth aspen

15

1

1

EH

Eastern hemlock

0

0.8916

1.012

GA

Green ash

13

1

1

GB

Gray birch

12

1

1

JP

Jack pine

0

0.916

1.01

NS

Norway spruce

0

0.8558

1.036

PB

Paper birch

0

0.8969

1.018

QA

Quaking aspen

0

0.8449

1.033

RM

Red maple

0

0.9214

1.012

RO

Northern red oak

11

1

1

RP

Red pine

0

0.928

0.999

RS

Red spruce

0

0.864

1.029

SB

Sweet birch

12

1

1

SM

Sugar maple

0

0.9383

1.006

TL

Tamarack (alias of TA in taper table)

0

1.511

0.8134

TA

Tamarack (eastern larch)

0

1.511

0.8134

WA

White ash

0

0.8834

1.019

WC

Northern white cedar

0

0.7797

1.057

NC

Non-commercial hardwood / northern white cedar fallback (context-dependent)

0

0.7797

1.057

WP

Eastern white pine

0

0.926

1

WS

White spruce

0

0.886

1.022

YB

Yellow birch

0

0.8688

1.028

OH

Other hardwood (group)

0

0.8923

1.019

OS

Other softwood (group)

0

0.8873

1.019

99

Other (unmatched-species fallback)

0

0.8898

1.019

B.5 Tree-level backgrou

nd mortality species parameters (section 5, tree.mort.spp)

Weiskittel et al. (2012). Tree survival is logistic in DBH and DBH^2 (b0, b1, b2); d.jump, scale, and shape parameterize the associated Weibull-form diameter-jump term. This table is broader than Appendix A (48 rows) because several species collapse to shared coefficient sets across related taxa (for example AH, BE, NM, SH, SV all share the generic hardwood-group coefficients).

Code

Common name

b0

b1

b2

d.jump

scale

shape

AB

American beech

2.153

-0.02698

0.0002203

41

20

4.5

AE

American elm

2.948

-0.1018

0.001828

22.5

40

2

AH

American hornbeam

4.552

-0.4627

0.0126

12

15

4.5

AP

Apple spp.

5.643

-0.4645

0.01327

16

10.6

4.1

BA

Black ash

1.718

0.0393

-0.002351

39

37

4.25

BC

Black cherry

4.863

-0.3696

0.01098

9.2

31

3.63

BE

Boxelder

4.552

-0.2313

0.0126

14

20

4.5

BF

Balsam fir

2.574

-0.08519

0.001597

53

40.6

1.51

BL

Black willow

-3.518

0.5008

-0.01144

10

10

2

BN

Butternut

9.614

-0.8619

0.02159

15

27.5

1.5

BO

Black oak

2.74

-0.04031

0.001485

40

40

2

BP

Balsam poplar

1.88

-0.3915

0.0298

33

33.6

4.75

BS

Black spruce

1.957

0.05354

-0.001038

34

22

3.75

BT

Bigtooth aspen

2.179

-0.01254

0.000853

40

30

3

BW

American basswood

-1.415

0.3205

-0.002971

15

30

3

CC

Choke cherry

4.863

-0.3696

0.01098

10

20

3.63

EC

Eastern cottonwood

-0.4585

0.1993

-0.002845

40.6

11

4.28

EH

Eastern hemlock

4.521

-0.06704

0.001204

50

40

3.5

GA

Green ash

7.206

-0.224

0.007037

34

39.6

3.41

GB

Gray birch

0.1923

0.1517

-0.003927

16

8.2

3.83

HH

Eastern hophornbeam

2.967

-0.101

0.007167

18

24

2

JP

Jack pine

-0.4488

0.194

-0.001954

30

20

3

NM

Norway maple

4.552

-0.2313

0.0126

24

40

4.5

NS

Norway spruce

17.48

-1.809

0.0617

15

35

2

OH

Other hardwood (group)

4.552

-0.5205

0.0126

12

10

4.5

PB

Paper birch

2.586

-0.05185

0.002185

26.4

41.2

1.88

PP

Pitch pine

12.16

-1.048

0.02331

25.8

41.6

4.41

PR

Pin cherry

-1.217

0.3211

-0.009715

10

40

2

QA

Quaking aspen

-0.4585

0.1993

-0.002845

60.6

11

4.28

RM

Red maple

2.167

0.05573

-0.001044

60.2

40.6

4.38

RO

Northern red oak

3.12

-0.04129

0.002298

41

40.4

3.27

RP

Red pine

1.136

0.1436

0.001844

30

30

3

RS

Red spruce

2.042

0.04257

-0.0004902

41

32

4.8

SB

Sweet birch

44.26

-2.425

0.1388

22

35.6

4.57

SH

Shagbark hickory

4.552

-0.2313

0.0126

15

10

4.5

SM

Sugar maple

2.707

0.008626

0.0007235

54.4

42

1.33

SP

Shortleaf pine (context) / scotch pine (bark table context)

5

-0.3

0.01

20

40

3

ST

Striped maple

4.552

-0.4627

0.0126

4.4

24.4

4.51

SV

Silver maple

4.552

-0.2313

0.0126

24

40

4.5

TA

Tamarack (eastern larch)

1.427

0.08863

-0.002123

30

33.6

4.5

WA

White ash

1.004

0.1654

-0.0005815

21

40

4

WC

Northern white cedar

3.648

-0.06067

0.0008508

40

45

5

WL

Willow spp.

-3.518

0.5008

-0.01144

10

10

2

WO

White oak

-4.864

0.6251

-0.006442

18

40

4

WP

Eastern white pine

3.338

-0.02945

0.0009562

61.2

41.2

2.87

WS

White spruce

0.5438

0.1052

0.0006333

19

54

2

YB

Yellow birch

2.697

-0.001251

0.0007521

48

60

2

99

Other (unmatched-species fallback)

2.697

-0.001251

0.0007521

48

60

2

B.6 Regional stand-mortality parameters (section 5, stand.mort.prob.region, stand.mort.ba.region)

Weiskittel et al. (2012), region-specific coefficients (ME, NB, NS, QC). Stand mortality probability: prob = exp(k)/(1+exp(k)), k = b0 + b1*BA* + *b2*BA^2 + b3*BAG* + *b4*QMD + b5*BA.BF* + *b6*BA.IH, evaluated against the region's cut threshold. Stand mortality basal area: BA.mort = (b0 + b1*pBA.bf* + *b2*pBA.ih) * BA^(b3 + b4*BAG); BF.mort* = *b5*BA.BF^b6 + b7*(QMD.BF/QMD).

Region

b0

b1

b2

b3

b4

b5

b6

cut

ME

0.6907

0.1492

-0.001856

-2.557

-0.05508

0.06415

0.04327

0.872

NB

0.6991

0.1251

-0.001856

-2.557

-0.05508

0.06415

0.04327

0.7268

NS

0.2757

0.1499

-0.001856

-2.557

-0.05508

0.06415

0.04327

0.9148

QC

1.047

0.1617

-0.001856

-2.557

-0.05508

0.06415

0.04327

0.7352

Region

b0

b1

b2

b3

b4

b5

b6

b7

ME

0.1858

0.2315

0.0202

0.5674

-2.037

0.06815

0.3345

0.09951

NB

0.5988

0.2315

0.0202

0.1889

-2.037

0.1461

0.3285

0.09951

NS

0.1302

0.2315

0.0202

0.5894

-2.037

0.08678

0.2244

0.09951

QC

0.1068

0.2315

0.0202

0.681

-2.037

0.01172

0.4941

0.09951

Thinning. Post-commercial-thinning diameter-increment modifier, Weiskittel, Hennigar, and Kershaw (2015) (the source code cites this reference; the main-text section 8 citation of Kuehne et al. 2016 for the same modifier should be reconciled against the code header before final publication). Defined only for BF and RS (all other species: modifier = 1). ddbh.thin.mod = 1 + exp(y0 + y1/((100*ba.perc.rmv*qmd.ratio) +0.01)) * y2^tst * y3^… (tst = years since thinning).

Species

y0

y1

y2

y3

BF

-0.2566

-22.7609

0.7745

1.0511

RS

-0.5010

-20.1147

0.8067

1.1905

Region

b1

b2

b3

b4

ME

-2.638

0.0114

-0.0076

0.0074

NB

-3.089

0.0071

-0.0037

0

Region

Code

b1

b2

b3

b4

b5

b6

b7

b8

ME

BF

-6.521

-0.4866

-0.0355

0.0316

1.509

-0.0175

0.0274

0.004

ME

BS

-6.521

-0.4866

-0.1231

0.0316

1.509

-0.0175

0.0274

0.0056

ME

RS

-6.521

-0.4866

-0.1231

0.0316

1.509

-0.0175

0.0274

0.0056

ME

WS

-6.521

-0.4866

-0.1755

0.0316

1.509

-0.0175

0.0274

0.0207

NB

BF

-6.831

0

-0.2285

0.2025

2.17

0

0

0.0029

NB

BS

-6.831

0

-0.2285

0.2025

2.081

0

0

0.0101

NB

RS

-6.831

0

-0.2285

0.2025

2.081

0

0

0.0101

NB

WS

-6.831

0

-0.2285

0.2025

1.58

0

0

0.0021

Code

stm.b0

stm.b1

stm.b2

lsw.b0

lsw.b1

lsw.b2

mst.b0

mst.b2

lf.b0

lf.b1

lf.b2

RO

-0.9491

0.0174

-0.2826

-1.114

-0.0322

0.791

-0.411

-0.5009

-4.068

0.0322

0.1139

SM

-0.9491

0.0174

0.7541

-1.114

-0.0322

-0.2325

-0.411

-1.135

-4.068

0.0322

0.6278

YB

-0.9491

0.0174

-0.0208

-1.114

-0.0322

0.298

-0.411

-0.7557

-4.068

0.0322

1.068

RM

-0.9491

0.0174

0

-1.114

-0.0322

0

-0.411

0

-4.068

0.0322

0

OH

0

0

0

0

0

0

0

0

0

0

0

Code

hr.b0

hr.b1

hr.b2

hr.b3

RO

-0.6886

-0.0001

-0.0184

-0.0393

SM

-0.6886

-0.0001

-0.1513

-0.0164

YB

-0.6886

-0.0001

-0.9851

0.0196

RM

-0.6886

-0.0001

0

0

OH

0

0

0

0

Code

b0

b1

QA

-2.791

0.0791

RM

-3.981

0.8343

RO

-0.7937

0.8944

YB

5.253

0.1528

PB

0

0

Li et al. (2011), ten sp.grp.ingrowth groups. Composition (count) model:

Group

b0

b1

b2

b3

b4

BCH

-2.564

0.002

2.662

-0.001

-0.0127

BF

-3.029

0.0027

2.778

0.0211

0.0221

RM

-0.6566

0.0123

1.767

-0.0421

-0.0283

SPR

-1.25

-0.0132

2.047

-0.0514

0.0351

WP

-5.107

-0.0117

3.882

0.0501

0.0726

OH

-2.983

-0.002

2.484

0.0673

-0.0167

OS

-4.718

0.007

3.227

0.1

0.0188

AB

-2.983

-0.002

2.484

0.0673

-0.0167

QA

-2.983

-0.002

2.484

0.0673

-0.0167

SM

-2.983

-0.002

2.484

0.0673

-0.0167

WC

-4.718

0.007

3.227

0.1

0.0188

Occurrence probability model parameters (two candidate models retained, GNLS and NLME; production defa

ult
note
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ops
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rowt
h.m
odel
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e):

Model

a1

a2

a3

a4

a5

a6

a7

b1

b2

b3

b4

b5

b6

b7

GNLS

-0.2116

-0.0255

-0.1396

-0.0054

0.0433

0.0409

0

3.898

-0.0257

-0.3668

0.0002

0.0216

-0.0514

0

NLME

-0.08217

0.1113

-1.24

-0.2319

0.03673

-0.7745

-0.1301

2.847

-0.03114

-0.2891

0.00335

0.2248

-0.08223

-0.03548

B.9 Taper species parameters (section 7)

Kozak variable-exponent taper (Li et al. 2012, conifers; Weiskittel and Li 2012, hardwoods), separate outside-bark (ob) and inside-bark (ib) coefficient sets where both exist; nomatch defaults to ib/OS.

Bark

Code

a0

a1

a2

b1

b2

b3

b4

b5

b6

b7

ob

AB

1.0694

0.9975

-0.012828

0.3921

-1.0546

0.77584

4.1035

0.1186

-1.0807

0

ob

BC

0.98022

0.99008

0.021502

0.60928

-0.54627

0.52219

1.6561

0.040879

-0.30281

0

ib

BF

0.88075

1.0149

0.019588

0.41952

-0.67233

0.5433

1.4818

0.064704

-0.34685

0

ob

BF

0.87046

0.99815

0.058482

0.30254

-0.60579

0.58886

0.88266

0.10328

-0.57433

0

ob

BP

1.0036

0.74425

0.28764

0.6634

-2.0048

0.7508

3.9248

0.027679

-0.13093

0

ob

BA

1.0036

0.74425

0.28764

0.6634

-2.0048

0.7508

3.9248

0.027679

-0.13093

0

ib

BS

0.80473

1.008

0.056011

0.35534

-0.4132

0.41527

1.1165

0.099017

-0.40992

0.11395

ob

BS

0.89638

0.97916

0.070704

0.28821

-0.30358

0.43523

0.28709

0.086104

-0.40775

0.37111

ob

BT

1.0201

1.0055

-0.011031

0.51045

-1.3264

0.55687

7.2108

0.07115

-0.57184

0

ib

EH

0.96024

1.0082

-0.025168

0.82526

1.9625

0.41523

-5.0616

0.0098395

-0.095533

0

ob

EH

0.84641

0.98432

0.080752

0.44544

-0.67147

0.50495

2.4894

0.12415

-0.72295

0

ob

GA

1.0852

1.1862

-0.22619

0.51988

1.4303

-0.34945

3.1953

0.13917

-0.29672

0

ob

GB

1.0264

0.88356

0.13075

0.61135

-0.11419

0.28832

2.6574

0.059005

-0.17513

0

ib

JP

0.93155

1.0082

-0.0041774

0.4313

-0.86367

0.5117

2.2325

0.059865

-0.3319

0.039631

ob

JP

0.84248

0.9928

0.073943

0.37222

-0.72323

0.45343

1.3375

0.073373

-0.31053

0.3964

ib

NS

0.93088

0.97361

0.035229

0.65078

-0.30356

0.37833

1.1882

0.031116

-0.031728

0

ob

NS

0.95095

0.99162

0.035718

0.50748

-0.44047

0.40586

1.285

0.014396

-0.078589

0.16973

ib

PB

0.71612

0.98112

0.13825

0.47822

0.30915

0.32663

-0.30206

0.085859

-0.27866

0

ob

PB

0.71612

0.98112

0.13825

0.47822

0.30915

0.32663

-0.30206

0.085859

-0.27866

0

ob

QA

0.5587

0.90478

0.30751

0.71313

-0.58835

0.4292

2.8516

0.038161

-0.13426

0

ib

RM

0.74583

1.0092

0.089093

0.58616

-0.86591

0.65392

3.0604

0.082762

-0.6486

0

ob

RM

0.74583

1.0092

0.089093

0.58616

-0.86591

0.65392

3.0604

0.082762

-0.6486

0

ob

RO

1.1751

1.0225

-0.069889

0.45057

-0.90288

0.58125

3.6267

0.16561

-1.1143

0

ib

RP

0.97179

1.0011

-0.015979

0.51143

-0.974

0.25844

4.7532

0.058462

-0.12372

0

ob

RP

1.0647

0.9949

-0.012383

0.45896

-1.0458

0.36145

4.0005

0.054337

-0.12803

0

ib

RS

0.89798

1.0058

0.016673

0.49501

-0.63375

0.38363

1.4138

0.08867

-0.29754

0.15192

ob

RS

0.88689

0.99543

0.054137

0.41116

-0.65802

0.41821

1.0911

0.10238

-0.40367

0.10484

ob

SB

0.84711

0.98754

0.076969

0.93226

-0.95458

0.48554

3.0295

0.076761

-0.2384

0

ob

SM

1.0517

0.9613

0.038604

0.85564

-0.24972

0.41494

1.2548

0.0413

-0.1135

0

ob

TL

0.76298

0.97932

0.12279

0.24594

-0.5649

0.66679

-0.072878

0.14365

-0.79119

0

ob

TA

0.76298

0.97932

0.12279

0.24594

-0.5649

0.66679

-0.072878

0.14365

-0.79119

0

ob

WA

0.85507

0.97689

0.077036

0.78191

-0.79176

0.4767

3.5004

0.085904

-0.48797

0

ib

WC

0.86119

0.98152

0.05682

0.40718

-0.054826

0.47809

-1.3251

0.15385

-0.53688

0

ib

NC

0.86119

0.98152

0.05682

0.40718

-0.054826

0.47809

-1.3251

0.15385

-0.53688

0

ob

WC

0.87698

0.97219

0.090503

0.31964

-0.49578

0.54661

-0.054012

0.13167

-0.45477

0

ob

NC

0.87698

0.97219

0.090503

0.31964

-0.49578

0.54661

-0.054012

0.13167

-0.45477

0

ib

WP

1.0488

1.0078

-0.045954

0.38085

-0.85956

0.34381

4.6084

0.11185

-0.55232

0

ob

WP

0.96198

0.98598

0.033318

0.38342

-0.75366

0.39253

3.4224

0.1006

-0.48562

0

ib

WS

0.75826

0.98482

0.099562

0.36505

-0.51501

0.55914

0.75846

0.070119

-0.44928

0.0783

ob

WS

0.72506

0.99993

0.11891

0.28603

-0.41705

0.58123

-0.56275

0.10138

-0.56377

0.096122

ob

YB

1.1264

0.94851

0.037132

0.76625

-0.028148

0.2334

4.857

0.075318

-0.20505

0

ob

OH

0.94721

0.97135

0.063182

0.63361

-0.54916

0.43901

3.1876

0.079154

-0.41278

0

ob

OS

0.88048

0.98853

0.066079

0.36555

-0.60725

0.48683

1.2824

0.09412

-0.44738

0

ib

OS

0.89648

1.0019

0.020707

0.39152

-0.39564

-0.011787

1.3351

0.076312

-0.28699

0

Code

th1

th2

a1

a2

gm1

gm2

phi

lmd

bet1

bet2

SM

6.579

0.0111

1.0682

1.1833

0.1031

0.2624

0.1516

1.1482

0.6637

3.0996

YB

7.5437

0.0103

0.9961

1.1042

0.1313

0.3539

0.2091

0.9478

0.5995

3.4205

PB

7.5437

0.0103

0.9961

1.1042

0.1313

0.3539

0.2091

0.9478

0.5995

3.4205

GB

7.5437

0.0103

0.9961

1.1042

0.1313

0.3539

0.2091

0.9478

0.5995

3.4205

SB

7.5437

0.0103

0.9961

1.1042

0.1313

0.3539

0.2091

0.9478

0.5995

3.4205

RM

7.5707

0.0105

1.5273

0.7684

0.0931

0.4223

0.1441

1.391

0.6453

4.0737

GA

3.3085

0.0276

1.2634

0.9088

0.1098

0.5198

0.184

1.7842

0.6719

5.1178

WA

3.3085

0.0276

1.2634

0.9088

0.1098

0.5198

0.184

1.7842

0.6719

5.1178

BT

3.3085

0.0276

1.2634

0.9088

0.1098

0.5198

0.184

1.7842

0.6719

5.1178

QA

3.3085

0.0276

1.2634

0.9088

0.1098

0.5198

0.184

1.7842

0.6719

5.1178

BP

3.3085

0.0276

1.2634

0.9088

0.1098

0.5198

0.184

1.7842

0.6719

5.1178

AB

8.9843

0.0107

0.7621

1.3734

0.0956

0.165

0.1924

1.2237

0.4626

1.0954

RO

12.834

0.0125

0.9038

1.095

0.0935

0.3971

0.2038

1.0457

0.5508

3.4681

BC

3.2042

0.0479

1.2507

0.8075

0.08

0.417

0.2227

2.7226

0.7065

4.6476

BF

5.3693

0.0171

1.4212

0.3003

0.089

0.6485

0.1916

1.8873

0.4764

2.6383

RS

6.8745

0.011

1.1241

0.4107

0.1376

0.4842

0.2038

1.2598

0.4986

2.7865

BS

6.8745

0.011

1.1241

0.4107

0.1376

0.4842

0.2038

1.2598

0.4986

2.7865

WS

6.8745

0.011

1.1241

0.4107

0.1376

0.4842

0.2038

1.2598

0.4986

2.7865

WP

7.1438

0.0123

0.8978

0.7872

0.0989

0.4985

0.2049

0.9247

0.5715

2.0482

TA

5.2913

0.0411

1.1291

0.6831

0.0745

0.5798

0.1896

1.5776

0.6616

6.0645

NS

5.2913

0.0411

1.1291

0.6831

0.0745

0.5798

0.1896

1.5776

0.6616

6.0645

JP

5.2913

0.0411

1.1291

0.6831

0.0745

0.5798

0.1896

1.5776

0.6616

6.0645

WC

5.4

0.0256

1.9295

0.8142

0.0943

0.9642

0.2761

1.8605

1.3432

1.3438

RP

7.6044

0.0148

1.2379

0.3304

0.0759

0.6611

0.3008

1.1569

0.5462

3.0627

EH

7.2442

0.0152

1.4008

0.8306

0.0856

0.4724

0.2011

1.5776

0.6616

6.0645

OH

9.0505

0.0241

1.298

0.7684

0.0684

0.4555

0.1769

1.6684

0.5408

4.1821

OS

6.4182

0.019586

1.3058

0.59379

0.093686

0.61553

0.22399

1.4626

0.68527

2.8423

99

7.115

0.016837

1.1484

0.85719

0.10123

0.46091

0.19561

1.3514

0.63404

3.3026

B.10 Honer volume species parameters (section 7)

Legacy total-tree cubic-volume equation, retained for cross-checking against the taper-integrated and NSVB-emulator volumes; columns p1.ht through p3.d are carried in the source table but not used in the current volume calculation.

Code

Common name

a

b

b2

WP

Eastern white pine

0.691

363.7

0.184

RP

Red pine

0.71

355.6

0.184

JP

Jack pine

0.897

348.5

0.184

TA

Tamarack (eastern larch)

0.897

348.5

0.184

LP

Loblolly pine (Honer volume table only, out of Acadian range)

0.694

343.9

0.184

BS

Black spruce

1.588

333.4

0.184

RS

Red spruce

1.226

315.8

0.184

WS

White spruce

1.44

342.2

0.184

BF

Balsam fir

2.139

301.6

0.184

NC

Non-commercial hardwood / northern white cedar fallback (context-dependent)

4.167

244.9

0.184

WC

Northern white cedar

4.167

244.9

0.184

EH

Eastern hemlock

1.112

350.1

0.184

QA

Quaking aspen

-0.312

436.7

0.184

BP

Balsam poplar

0.42

394.6

0.184

BT

Bigtooth aspen

0.42

394.6

0.184

PB

Paper birch

2.222

300.4

0.184

GB

Gray birch

2.222

300.4

0.184

SB

Sweet birch

2.222

300.4

0.184

YB

Yellow birch

1.449

344.8

0.184

RM

Red maple

1.046

384

0.184

SM

Sugar maple

1.046

384

0.184

BW

American basswood

0.948

401.5

0.184

AB

American beech

0.959

334.8

0.184

WA

White ash

0.959

334.8

0.184

BC

Black cherry

0.033

393.3

0.184

RO

Northern red oak

1.512

336.5

0.184

OH

Other hardwood (group)

1.512

336.5

0.184